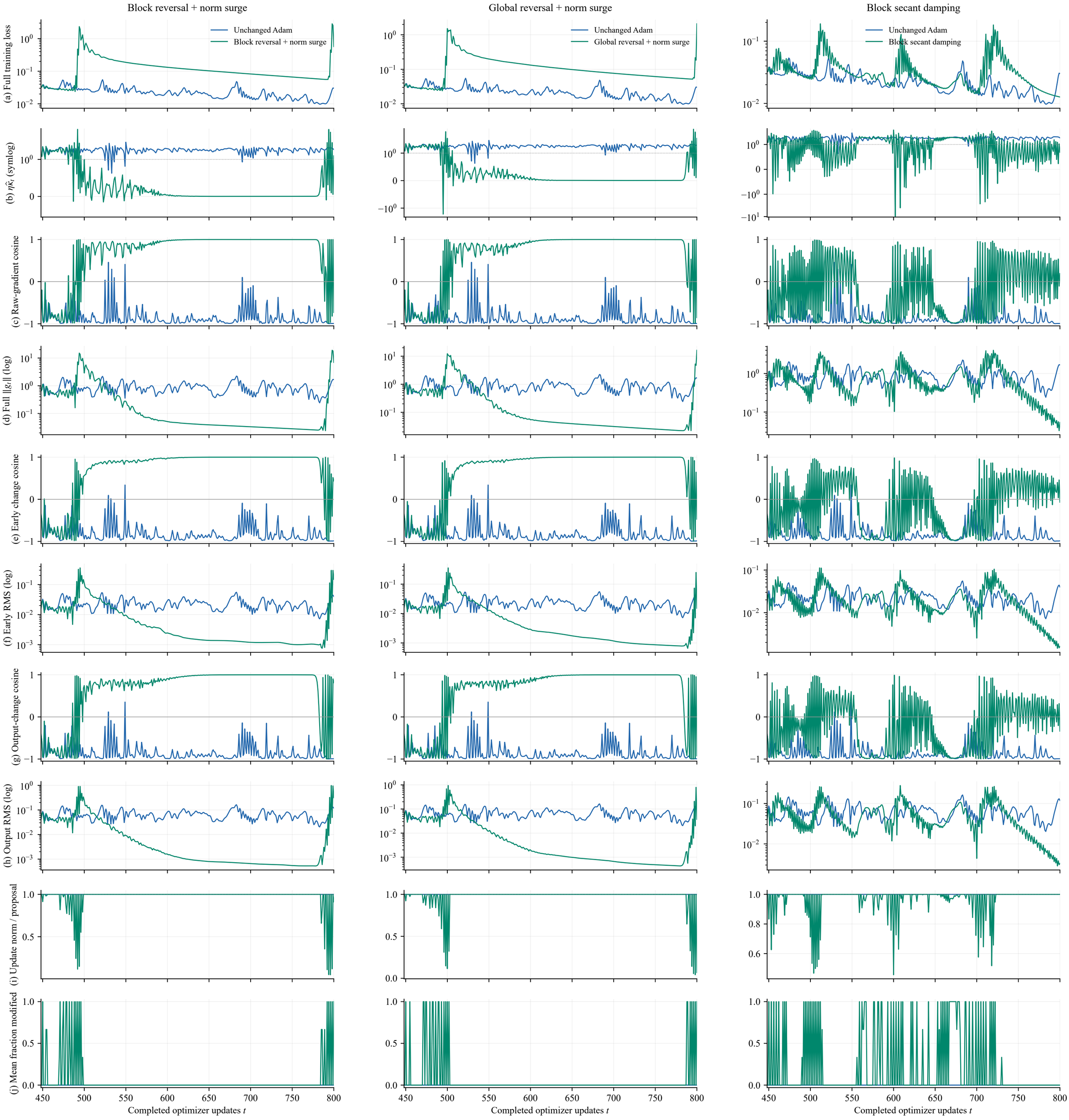
MLP: transient diagnostic-triggered damping fails

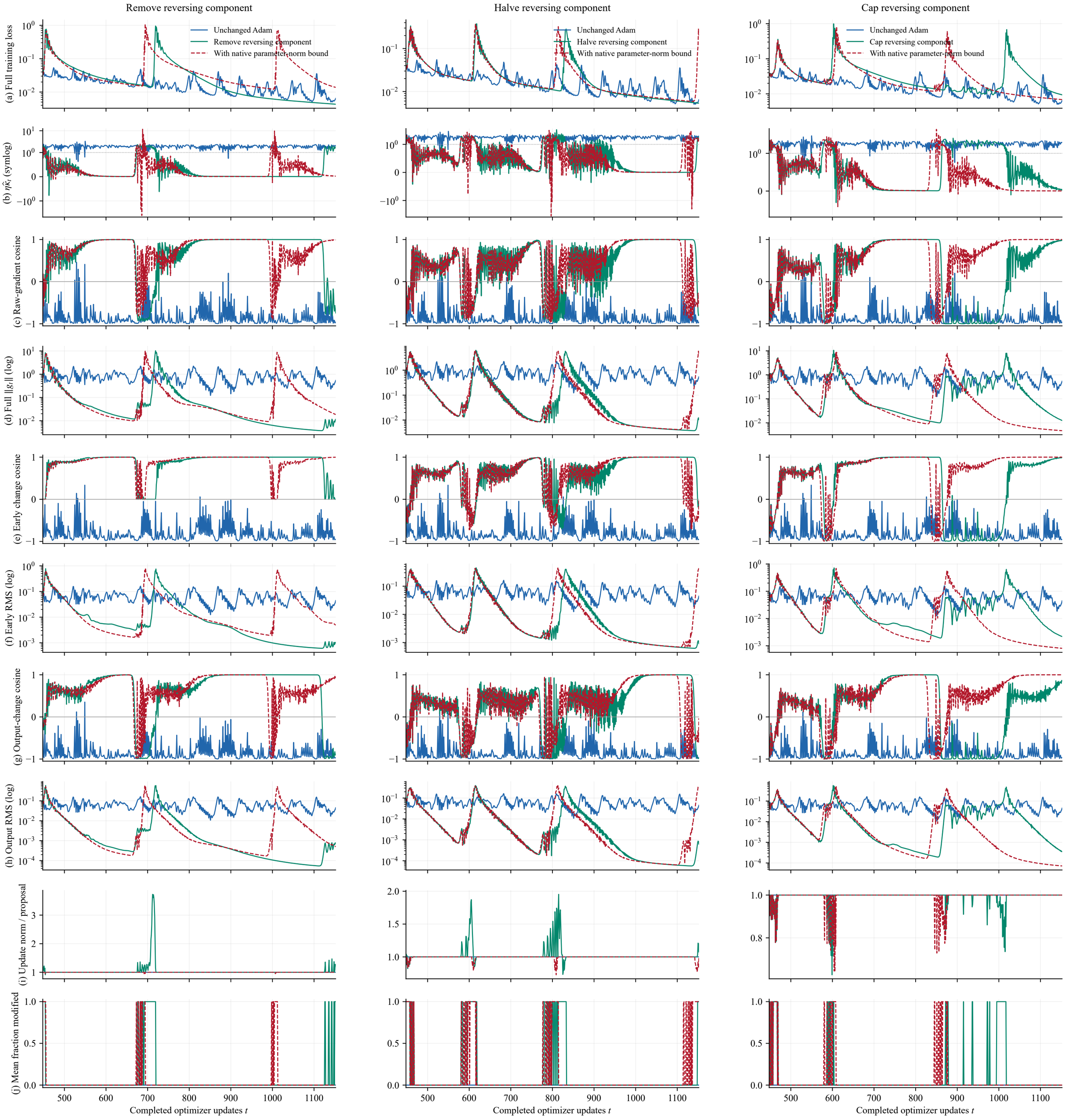
Loss uses log axes to retain the untreated trajectory beside the much larger intervention excursions.



Full objectives; fixed diagnostic probes; outgoing representation changes; no smoothing. Each column uses its own unchanged baseline.
Early: MLP first tanh, ResNet conv1 output, Transformer first attention output. Loss and magnitude scales are independent by column.
The modified-fraction row averages targeted coordinate/neuron/block fractions; this is not a whole-model parameter fraction.
Undefined cosines and zero values on logarithmic axes appear as gaps.

MLP: past-axis damping, with and without a parameter-norm bound

Loss uses log axes. Fewer reversals and better endpoints can coexist with much larger transient loss.



Full objectives; fixed diagnostic probes; outgoing representation changes; no smoothing. Each column uses its own unchanged baseline.
Early is the first affine projection. Only its negative component along the preceding actual affine change is modified.
Last row is 1 when that component is modified. This uses one preceding direction, not a future-fitted SVD basis; all other increments stay native.
Undefined cosines and zero values on logarithmic axes appear as gaps.

MLP: movement budgets and the timing of release

Backtrack the complete Adam increment until actual fixed-probe displacement RMS meets its prefix budget.

Output movement budget (q50)

All tracked movements (q50)

Output movement budget (q75)

Unchanged Adam

Output movement budget (q50)

All tracked movements (q50)

Output movement budget (q75)

With slow recovery

(a) Full training loss

(b) $\eta\hat{\kappa}_t$ (symlog)

(c) Raw-gradient cosine

(d) Full $\|g_t\|$ (log)

(e) Early change cosine

(f) Early RMS (log)

(g) Output-change cosine

(h) Output RMS (log)

(i) Update norm / proposal

(j) Mean fraction modified

Completed optimizer updates t

Full objectives; fixed diagnostic probes; outgoing representation changes; no smoothing. Each column uses its own unchanged baseline.
Early: MLP first tanh, ResNet conv1 output, Transformer first attention output. Loss and magnitude scales are independent by column.
The modified-fraction row averages targeted coordinate/neuron/block fractions; this is not a whole-model parameter fraction.
Undefined cosines and zero values on logarithmic axes appear as gaps.

ResNet: diagnostic damping, uniform scaling, and distributed caps

Compare a blockwise reversal/norm rule, half increments, and prefix-q75 caps across all 31 convolutions.

Block reversal + norm surge

Uniform half increments

Distributed displacement cap (q75)

(a) Full training loss

Unchanged Adam

Block reversal + norm surge

Uniform half increments

Distributed displacement cap (q75)

(b) $\hat{\eta}\hat{\kappa}_t$ (symlog)

(c) Raw-gradient cosine

(d) Full $\|g_t\|$ (log)

(e) Early change cosine

(f) Early RMS (log)

(g) Output-change cosine

(h) Output RMS (log)

(i) Update norm / proposal

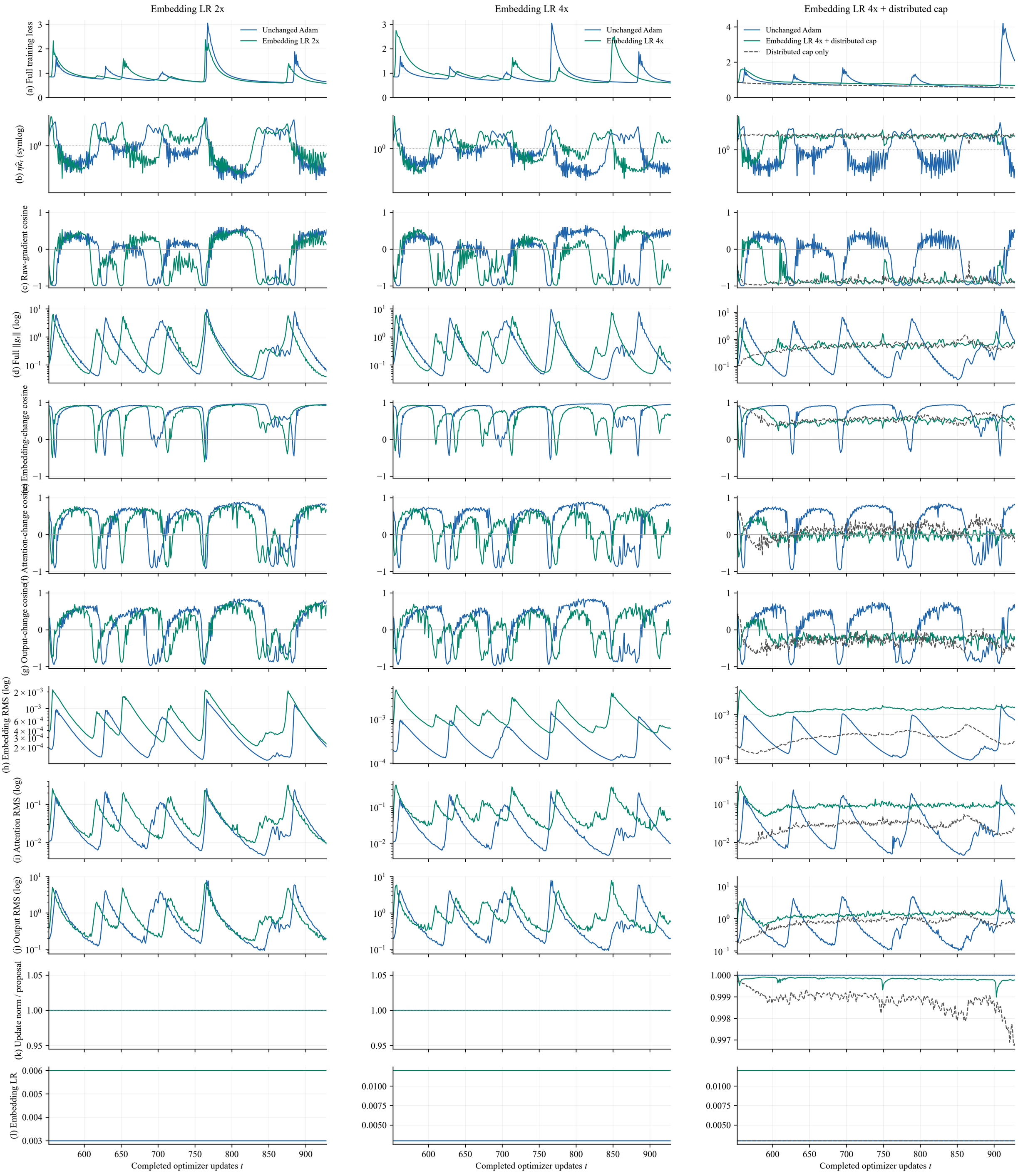
(j) Mean fraction modified

Completed optimizer updates t

Full objectives; fixed diagnostic probes; outgoing representation changes; no smoothing. Each column uses its own unchanged baseline.
Early: MLP first tanh, ResNet conv1 output, Transformer first attention output. Loss and magnitude scales are independent by column.
The modified-fraction row averages targeted coordinate/neuron/block fractions; this is not a whole-model parameter fraction.
Undefined cosines and zero values on logarithmic axes appear as gaps.

Transformer: larger embedding learning rates

All supervised tokens retained; fixed 64-sequence probe. The combined intervention also caps both QKV and feed-forward affines.



Full objectives; fixed diagnostic probes; outgoing representation changes; no smoothing. Each column uses its own unchanged baseline. Embedding and decoder are untied. Base LR 0.003; only embedding LR is multiplied. Update-norm ratio is relative to that native proposal. Columns 1/2 use RTX 4090; column 3 uses RTX 5090. Their control trajectories differ; each effect is compared within its own worker. Undefined cosines and zero values on logarithmic axes appear as gaps.